Beilong Tang

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EDUCATION

North Carolina State University (NCSU) Aug. 2025 – May 2030
Ph.D. in Electrical Engineering Raleigh, USA
- GPA: 4.0/4.0

Duke University & Duke Kunshan University (DKU) Dual Degree Aug. 2021 – May 2025
B.S. in Applied Math & Computational Sciences; Computer Science (Duke) Durham, USA
B.S. in Computer Science (DKU) Kunshan, China
- GPA: 3.886/4.0
- Graduated with *Signature Work with Distinction* and *Magna Cum Laude* (top 15% GPA) award
- Thesis: *Utilizing discrete tokens and language models for target speaker extraction*

RESEARCH EXPERIENCE

Research Assistant for Generative Models Aug. 2025 – Present
The *VISSTAL* Lab, NCSU Supervisor: Dr. Hamid Krim
- Design geometry-aware diffusion processes that respect the manifold structure of data and latent spaces for high-quality image generation.
- Proposed Polar Diffusion, which decouples radial and tangential denoising in diffusion models (first author, submitted to ICLR 2027).
- Developed a spherical latent diffusion framework that defines the diffusion process directly on the hypersphere of latent embeddings, outperforming flow matching baselines.
- Managed distributed multi-GPU training of diffusion models on a GPU cluster.

Research Assistant for Speech Separation and Enhancement May 2024 – June. 2025
The SMIIP Lab, DKU Supervisor: Dr. Ming Li
- Developed LauraTSE and TSELM, target speaker extraction systems built on discrete speech tokens and auto-regressive decoder-only language models (e.g., LauraGPT).
- Investigated discretization methods for mixed (multi-speaker) speech.
- Proposed SEF-MK, a speaker-embedding-free voice anonymization method based on multi-k-means quantization.

Research Assistant for Federated Learning Project May 2023 – Jan. 2024
The Edge Intelligence Lab, DKU Supervisor: Dr. Bing Luo
- Deployed Federated Learning (FL) and Federated Analytics (FA) on edge devices.
- Built a privacy-preserving health monitoring system using FL and FA, deployed across the DKU campus (FedCampus).
- Developed full-stack mobile applications for Android and iOS, including the cross-platform FL framework FedKit.

PUBLICATIONS

Tang, B., Bao, Y., & Krim, H. (2026). Polar Diffusion: Decoupling Radial and Tangential Denoising in Diffusion Models. *Under review at ICLR 2027.*

Zeng, B., **Tang, B.**, Xiang, W., & Li, M. (2026). Discriminative-Generative Target Speaker Extraction with Decoder-Only Language Models. *arXiv preprint arXiv:2601.06006.*

Tang, B., Zeng, B., & Li, M. (2025). LauraTSE: Target Speaker Extraction using Auto-Regressive Decoder-Only Language Models. *IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*.

Tang, B., Miao, X., Wang, X., & Li, M. (2025). SEF-MK: Speaker-Embedding-Free Voice Anonymization through Multi-k-means Quantization. *IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*.

Lin, Y., Cheng, M., Li, Z., **Tang, B.**, & Li, M. (2025). Diarization-Aware Multi-Speaker Automatic Speech Recognition via Large Language Models. *arXiv preprint arXiv:2506.05796*.

Tang, B., Zeng, B., & Li, M. (2024). TSELM: Target Speaker Extraction using Discrete Tokens and Language Models. *arXiv preprint arXiv:2409.07841*.

Geng, J., **Tang, B.**, Zhang, B., Shao, J., & Luo, B. (2024). Demo: FedCampus: A real-world privacy-preserving mobile application for smart campus via federated learning & analytics. In *Proceedings of the 25th International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing (MobiHoc '24)*. ACM.

He, S., **Tang, B.**, Zhang, B., Shao, J., Ouyang, X., Nugraha, D. N., & Luo, B. (2024). FedKit: Enabling cross-platform federated learning for Android and iOS. In *IEEE INFOCOM 2024 - IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS)* (pp. 1–2). https://doi.org/10.1109/INFOCOMWKSHPS61880.2024.10620662

ACADEMIC SERVICE & TEACHING

Reviewer for ICASSP 2027
Reviewer for ICME 2025, 2026

Teaching Assistant for ECE 514: Random Processes NCSU
Teaching Assistant for ECE 411: Intro to Machine Learning NCSU
Teaching Assistant for COMPSCI 201: Intro to Programming and Data Structures DKU

AWARDS

- DKU Undergraduate Graduation with Distinction – 2025
- DKU Signature Work with Distinction – 2025
- DKU Magna Cum Laude (top 15% GPA) award
- Dean's List (top 30% academic performance per term) and Dean's List with Distinction (top 10%)
  - Dean's List: Fall 2021, Fall 2022
  - Dean's List with Distinction: Spring 2022, Spring 2023, Fall 2024, Spring 2025
- 100% Tuition Scholarship for Undergraduate Program: UGRD Entrance Scholarship, JF Progressive Scholarship
- Summer Research Scholarship (SRS) 2023 & 2024
- Silver Medal, International Genetically Engineered Machine (iGEM) 2023 DKU Team Nov. 2023

SIDE PROJECTS

Research Assistant for Search Engine Research Project Oct. 2022 – May 2023
The Search So Team, DKU Instructor: Dr. Jiang Long

- Developed benchmarks to evaluate the performance of our self-developed search engine.
- Created DKU-Wiki in Vue and Django to demonstrate the search engine's functionality.

Team Lead & Developer for Student Entrepreneurship Project Oct. 2022 – Jan. 2023
The Roomie Student Entrepreneurship Team, DKU

- Developed an app that matches college students with roommates based on MBTI personality types.
- Collaborated with the leadership team to drive product vision and execution.

Open-Source Developer for Repolab-Quiz May 2022 – July 2022
DKU Instructor: Dr. Jiang Long

- Independently developed a dynamic quiz website using Django that supports fill-in-the-blank questions, multiple-choice questions, and Java programming questions.
- Used as teaching material for COMPSCI 201.

SERVICES & LEADERSHIP

- President for DKU Running Club May 2024 – Aug. 2025
  - Led daily practices.
  - Collaborated with fellow club leaders to build a welcoming and inclusive running community.
  - Competed in cross-country races, half marathons, and marathons.
- Vice President for DKU Basketball Club & DKU Basketball Varsity Team Member May 2021 – Jan. 2024
  - Organized and coordinated club basketball events.
  - Managed financial planning and budgeting for club events and activities.

SKILLS & HOBBIES

- **Languages**: English (fluent) with TOEFL score of 107, Mandarin (native)
- **Programming Languages**: Python, Java, JavaScript, SML/NJ
- **Frameworks and Tools**: PyTorch, Django, Vue, MySQL
- **Hobbies**: Running, Basketball, Free Boxing, Drumming